

WIRELESS CONTROL Of ROBOTIC CAR Through MATLAB GUI

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This project presents the implementation of a MATLAB graphical user interface (GUI)-based approach to wirelessly control the movement of a robotic car. Commands to move the car in forward, reverse, right and left directions are sent from the GUI, and processed by Arduino present on transmitter side. After processing the command(s), a code is sent wirelessly to the receiver.

The receiver side decodes the code and makes the robot move accordingly. MATLAB-based GUI for wireless control of the robot is shown in Fig. 1. Prototypes for the transmitter and the robotic car (present on receiver side) are shown in Figs 2 and 3, respectively.



Fig. 1: MATLAB GUI for controlling the wireless robot

Circuit and working

Block diagrams of transmitter and receiver sides are shown in Figs 4 and 5, respectively.

Components used in this project are as follows:

Arduino Uno board. Arduino Uno is an AVR ATmega328P microcontroller (MCU)-based development board with six analogue input pins and 14 digital I/O pins. The MCU has 32kB ISP flash memory, 2kB RAM and 1kB EEPROM. The board provides the capability of serial communication via UART, SPI and I2C. The MCU can operate at a clock frequency of 16MHz. In this project, digital I/O pins 9, 10, 11 and 12 of Arduino are configured as output pins.

HT12E. The 2^{12} series encoders are generally used for remote control system applications. These encoders are capable of encoding information that consists of n address bits and $12-n$ data bits. Each address/data input can be set to one of the two logic states. Programmed address/data are transmitted together with header bits via an RF or IR transmission medium upon receipt of a trigger signal.

HT12D. The 2^{12} decoders are a series of CMOS LSIs for remote control

system applications. These are paired with HT12E encoders. Decoders receive serial addresses and data from programmed 2^{12} series of encoders that are transmitted by a carrier using an RF or IR transmission medium.

The decoders compare serial input data three times continuously with their local addresses. If no error or matched codes are found, input data codes are decoded and then transferred to output pins. The 2^{12} decoders are capable of decoding information that consists of n bits of addresses and $12-n$ bits of data. For a valid transmission, address bits of both encoder and decoder must be the same.

434MHz TX-RX modules. These 434MHz transmitter (TX) and receiver (RX) modules are electronic devices used for transmitting and receiving RF signals between two devices. These are used widely in electronic design, owing to the difficulty of designing



Fig. 4: Block diagram of transmitter side

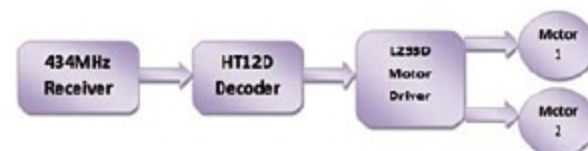


Fig. 5: Block diagram of receiver side

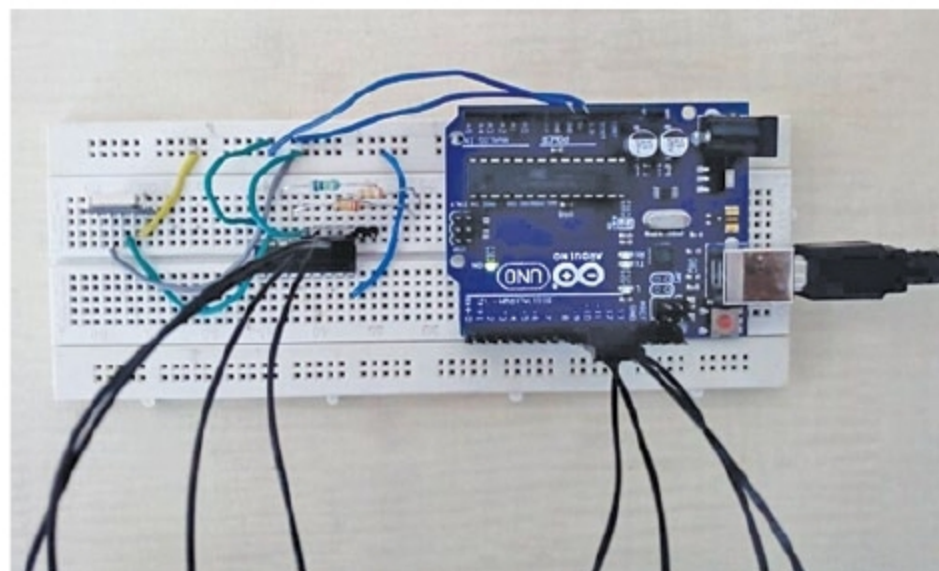


Fig. 2: Authors' prototype of transmitter side

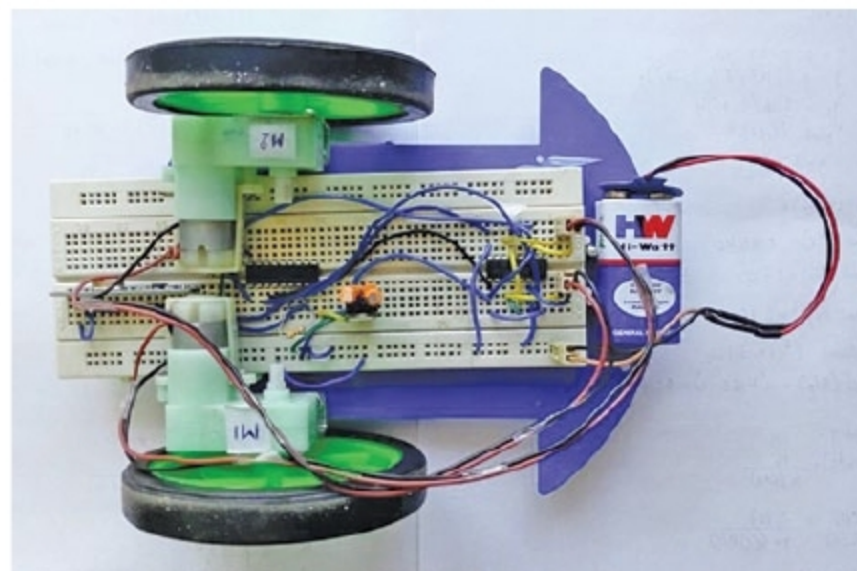


Fig. 3: Authors' prototype of robotic car (receiver side)